

TOMMASO AIELLO

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Education

- **Bocconi University:**
 - **Master's** in *Artificial Intelligence* 2025-2027
- Exchange program at *University of Maryland* Average: 28.6/30
 - **Bachelor's** in *Applied Mathematics and Computational Sciences for AI* 2022 - 2025
- Exchange program at *City University of Hong Kong* Average: 29.1/30

Research Experience

- **Statistical Physics and Diffusion Processes**
Bocconi University,
2025 - Current
Conducting theoretical research on diffusion processes in mean-field statistical-physics models, using analytical methods from probability and statistical mechanics.
Aimed at deriving mathematical results on phase selection, transition timescales and stability in Curie-Weiss and hierarchical systems.
- **Bachelor's Thesis**
Bocconi University,
Milan, Italy
October '24 - July '25
Built a Transformer-based Denoiser model to analyze how hierarchical data structures impacts diffusion model performance, highlighting key phase transitions in learning capacity of architectures based on sparsity and complexity of data, generated systematically in a controlled environment .

Industrial Experience

- **Data Science Intern**
BNL BNP Paribas
Rome, Italy
July - Sep 2024
Developed a Net Promoter Score prediction model using Markov-chain regression and performed statistical analysis of client profiles and satisfaction patterns using Python and SQL, alongside interactive Power BI dashboards to monitor financial assets and relationship-manager performance.

Selected Projects

- **Pydeia** - 1st Place at
"InnovAlte" Agentic-AI
Hackathon
Built a conversational AI assistant helping high-school students identify suitable university programs. Using a tailor-made recommendation system based on semantic analysis, this project dialogues with the Agentic AI framework provided by the hosts.
- Diffusion Processes for
Generative Hierarchical
Models
Built a modular framework for generating hierarchical sequences, training Transformer denoisers and analysing reconstruction across different structures and noise schedules focusing on phase transitions in reconstruction quality.
- Two-View 3D Reconstruction
Implemented modular camera calibration, SIFT feature matching, eight-point estimation, epipolar geometry and triangulation for Computer Vision Applications

Skills & Additional Information

- Programming: Python, SQL, R, C, C++ and JavaScript
- Languages: Italian (native); English (C1); French and Spanish (intermediate)
- Tools: PyTorch, pandas, OpenCV, Power BI, Git
- Full-tuition scholarship from Bocconi University.
- Regional math's Olympiads finalist at 15.
- Bass guitar player & Jazz enthusiast.
- Passionate swimmer

